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Construction of Personal Health Knowledge Graphs for Clinical Data Harmonization in Breast Cancer

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Article

Keywords: breast cancer, clinical registry, data harmonization, knowledge graph, artificial intelligence

Posted Date: February 9th, 2026

DOI: https://doi.org/10.21203/rs.3.rs-8404311/v1

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Additional Declarations: Competing interest reported. I.Z. is employed by b!loba. K.S., T.P., and S.B. are employed by Ontotext AD. The remaining authors declare no competing interests.

Abstract

Background

Personal health data contain valuable information for breast cancer management. Integration of heterogeneous data and maintenance of clinical registries are time-consuming and labor-intensive. We aimed to leverage an Artificial Intelligence (AI)-powered virtual assistant supporting semi-automated curation, including data quality enhancement, and publishing of personal health data. Identifiable data of breast cancer patients can be transformed into interoperable personal health knowledge graphs for secondary use.

Methods

With patient-informed consent, breast cancer patient data were extracted from the hospital systems and ingested into the virtual assistant. Data items were mapped and transformed into target concepts within a knowledge graph compliant with a reference ontology. Integrated classic and AI tools were used to support transformation of individual patients' data into a personal health knowledge graph (PHKG). Each graph was assessed by a Shapes Constraint Language (SHACL)-based validator to ensure the data quality. An RDF Query Language (SPARQL) query was executed on top of validated PHKGs from multiple patients to extract the relevant data elements and generate a local breast cancer registry, interoperable with registries generated in the same way across three different hospitals.

Results

The first version of the AI-powered virtual assistant prototype was developed and deployed in our hospital, as well as in two other hospitals in Austria and in Estonia. Twelve tables with 184 data items, including structured, semi-structured and fully narrative elements, were extracted from the local hospital systems. Data categories included demographics, diagnosis, medical history, pathological reports, laboratory tests, surgical records, therapy, and follow-up after previous treatments. Personal health knowledge graphs incorporating the data elements required for the Breast Cancer (BC) registry were constructed after data transformation. A SPARQL query was subsequently developed to build a local BC registry that automatically retrieved these relevant elements. The same approach took place in the other two hospitals.

Conclusion

The proposed workflow of semi-automated health data curation and quality enhancement from heterogeneous data sources to interoperable and reusable output is feasible. It provides a potential solution to enhance medical data interoperability and facilitate the maintenance of clinical registries.

Introduction

Breast cancer, despite the annual reduction in incidence and mortality rates in high-income countries, remains the leading cancer diagnosis and cause of death in females worldwide[1]. Unfortunately, the burden of breast cancer is unequally distributed, as lower developed countries report high mortality rates, despite lower incidence rates than very highly developed countries[2]. Treatment regimens for patients with breast cancer include a combination of mastectomy, radiation therapy, chemotherapy, hormonal therapy, immunotherapy and targeted therapy, depending on different types and stages of diseases[3]. Current challenges in breast cancer care are the high prevalence and need for early diagnosis[1]. In addition, improving the quality of life of breast cancer patients is another challenge, due to adverse events and toxicities after different treatments. To address those challenges, personalized medicine has been advocated. Personal health data can be a rich resource to realize personalized medicine because these data can facilitate earlier screening opportunities, more accurate diagnosis, individualized treatment and research[4]. In addition, personal health data of breast cancer patients can be used to develop clinical registries. High-quality cancer registries contain important information about disease causes, treatment performance, care outcomes and trends in cancer survival[5] and are resources for cancer monitoring and control, assessment of quality of care, and clinical research[6].

The value of cancer registries highly relies on the quality of data[7]. Completeness, timeliness, comparability, and validity are the four key dimensions of data quality in the registries[8]. Personal health data to be collected in cancer registries usually come from hospital electronic health records (EHRs). However, these data are characterized by multi-modalities, heterogeneity and often a lack of structure, which affects their interoperability and reuse for further applications[9]. To ensure the quality of a clinical registry, the data extraction and transformation process (data curation process) from heterogeneous data sources to standardized data elements must be harmonized. However, the establishment and maintenance of data quality in cancer registries is currently often labor-intensive. The process of describing and coding of cancer cases in the data sources represents the main challenge to establish a cancer registry, requiring extensive human and other resources. Additionally, variability across human annotators may generate inconsistencies. Consequently, the accuracy and validity of the data curation process cannot be guaranteed[10].

Linking cancer registries may save time and costs and can further increase the potential use of health data. However, registry linkage is often difficult as the data processing in different registries involves handling different standards. To enable linkage of cancer registry, an effective and harmonized method to curate data is required.

To address these challenges in personal health data collection, artificial intelligence methods, in particular Natural Language Processing (NLP)[11], Entity Linking (EL)[12] and deep learning (DL)[13], have been explored and applied to automate health data curation and harmonization. EL methods and Large Language Models (LLMs) have also been used in the development of medical knowledge graphs[13–15].

Knowledge graphs have been widely used in integrating heterogeneous data and representing domain knowledge[16]. A knowledge graph is defined as a graph structure that represents the relationships among

domain entities or events from the real-world, accompanied by domain rules and restrictions. The development of knowledge graphs relies on a pre-defined ontology which integrates several terminologies to indicate the relationships between different data items. The Resource Description Framework (RDF) is a common standard used for representing data relationships. As its main concept, RDF triples form the basic elements of knowledge graphs[17]. The SPARQL language is typically used to query the RDF triples and graph structure for extracting targeted information[18]. Therefore, individual health information can be curated into personal health knowledge graphs (PHKGs) to represent comprehensive medical information of individuals. For instance, a PHKG can capture all relevant medical information for a given patient, including their medical history, symptoms, diagnosis, treatments, and clinical outcomes[19,20].

Even though general and personal health knowledge graphs have been studied in disease diagnosis, genomics, drug discovery, and support for treatment related decision-making[21], their applications in breast cancer are still under development. One study combined electronic medical records, clinical guidelines, and medical encyclopedia to develop knowledge graphs for diagnosis support[22]. Another study extracted features from mammograph text reports and constructed a knowledge graph for breast cancer diagnosis[23]. *Jin et al.* integrated data from literature, constructing a knowledge graph for clinical decision-making in breast cancer[3]. Knowledge graphs have also been developed to reveal specific gene assays[24], represent risk factors[25] or discover biomarkers in breast cancer[26]. These studies were mainly conducted in a single center, based on literature and medical records, without evaluating the performance of workflow for the construction of PHKGs under a prospective context. Moreover, the developing process requires substantial manual efforts, especially in ontology definition and updates, data mapping, quality control and graph maintenance according to new data. As such, knowledge graphs in healthcare have potential but still require further research to determine their value.

In this study, we propose an innovative workflow driven by an AI-powered virtual assistant prototype to realize semi-automated curation of identifiable personal health data from electronic medical records into PHKGs, including core data elements that populate a BC registry. Relevant data elements were then queried from the PHKGs to form a site-specific BC registry. This study is a use case in the scope of the AI powered DAta curation and Publishing Virtual Assistant (AIDAVA) project.

The main objective of the AIDAVA project is to develop and test an intelligent virtual assistant (VA) prototype that integrates multiple data processing techniques, including AI and classical extract, transformation and load (ETL), then assists in structuring heterogeneous personal health data into the interoperable format as PHKGs. Data in the PHKGs can be published and reused for diverse clinical applications[11]. The prototype is tested and evaluated in four clinical sites across three European countries, including Maastro Clinic and Maastricht University Medical Centre (MUMC+) in the Netherlands, North Estonia Medical Centre (NEMC) in Estonia and Medical University of Graz (MUG) in Austria. Disease focuses are breast cancer and myocardial infarction[27].

Methods and Materials

Workflow Overview

The AIDAVA VA prototype is being implemented in two-step generations. The proposed Generation I (G1) prototype includes an interactive platform as the front-end for data ingestion, publishing and human-machine interaction, orchestrating intelligent tools for data curation at the back end. Several AI-based tools, including Optical Character Recognition (OCR), multi-lingual NLP, DL and EL are used as components of the workflow to realize semi-automated data curation into PHKGs, data publishing, reuse and quality enhancement. We conducted this pilot study in Maastro Clinic as a site-specific use case for breast cancer (Fig. 1 and Fig. 2).

After the patients provided informed consent, data collection ensued in Maastro Clinic between October 2024 and December 2024. The workflow of data curation and quality enhancement includes three main phases[28] (Fig. 3).

Data Source Onboarding

Stage 1 Data Source Documentation

Documentation includes the creation of source level and technical level metadata. Source level metadata contain information about the data origins, descriptions and the methods of data generation. Technical metadata (e.g. data transfer specification files (DTS)) relate to data formats, attributes and range restrictions. The metadata files were stored in a dedicated catalogue to maximize semi-automated curation. They were used as references each time when the AIDAVA VA prototype ingested patient data.

Stage 2 Preparations for data mapping

2.1 Ontology development

A reference ontology was developed as the standard for data mapping and modelling. To ensure that the PHKGs can effectively represent breast cancer data elements, this study adopted a top-down approach to define the entities and frameworks of the reference ontology at the concept level. It was developed by integrating widely used medical terminologies such as SNOMED CT[29] and ICD-10[30]. Besides medical terms, the reference ontology is compliant with the international Health Level Seven Fast Healthcare Interoperability Resources (HL7 FHIR)[31], therefore supports automatic generation of the International Patient Summary (IPS) from the PHKG for each patient. IPS belongs to one of the six critical categories of European Health Data Space (EHDS)[32]. The Swiss Personal Health Network framework was used as the basic schema of the reference ontology[11]. Extensions have been made based on the personal health data structure along with the domain expertise from a team of experts in ontology and clinical medicine.

2.2 Data transformation and mapping settings

Mapping rules and compatible data harmonization were defined according to the reference ontology. Patient data cleaning and further transformation were conducted by using an adapted open-source tool

OpenRefine (https://github.com/ensaremirerol/aidava-openrefine/tree/main). Transformation actions targeted certain structured data types (e.g. datetime, unit, frequency, etc.) (Fig. 4). An entity linking module was developed to trigger automated extraction of text snippets in the structured data and coding into target concepts. Clinical concepts were extracted from full narrative texts and standardized by an NLP tool, according to SNOMED CT.

Manually created synthetic data were used to preliminarily test the curation workflow. After transformation, the data were imported into the internally developed mapping tool (https://github.com/MaastrichtU-IDS/RDFCraft), then mapped into target concepts with relationships defined in the reference ontology to construct the basic structure of PHKGs. RDF triples were generated as the graph outputs, then validated by a SHACL engine. The SHACL rules define constraints to identify any conflicts between the constructed graph and the mapping rules defined by the reference ontology, such as inconsistent relationships against the ontology, coding errors, invalid and/or missing values. Valid PHKG outputs were loaded into a graph database (GraphDB, Ontotext, Sofia, Bulgaria).

Data Source Curation and Quality Enhancement

After consent was obtained, the data of breast cancer patients extracted from the clinical records were structured into documents and ingested to the AIDAVA VA prototype. Once the list of documents was shown to the patient at the prototype front end, the curation workflow was triggered at the back end to transform the data into a PHKG for each patient. Next, the PHKG as RDF triples were validated using SHACL for the quality enhancement process. In case of any conflicts detected, the system would raise a dialogue at the user interface to ask for support by a human-in-the-loop (HITL) process. These HITL dialogues were addressed to the patients or data curators based on their medical literacy and digital skill levels. Stakeholders completed an internally developed questionnaire designed specifically for this project to evaluate their medical literacy and digital skill levels (questionnaire example in the GitHub repository). The underlying HITL process was designed to support manual efforts for data validation, data correction and quality improvement. Upon successful validation and curation, the PHKG for each patient, including real patient data, was stored in the GraphDB database for subsequent analysis and knowledge discovery.

Data publishing and secondary use

We have developed information queries to retrieve patient information and support data reuse using the SPARQL language over PHKGs. The first implementation for data publishing is to generate the International Patient Summary (IPS), showing comprehensive information on the patient's medical records. The second query focuses on extracting target data elements from PHKGs to form a local breast cancer registry that is consistent with the standard of the Dutch National Cancer Registry (NCR)[33] and the NABON Breast Cancer Audit (NBCA)[34]. With the same ontology and therefore PHKGs available across the AIDAVA medical sites, a federated query can be developed by filtering the common data elements from all sites, demonstrating interoperability on multicenter heterogeneous data.

Results

Data characteristics

Identifiable data of fourteen breast cancer patients were extracted from Maastro Clinic's EHR system. For each patient, 184 data items were organized into twelve table sources that covered eight categories as follows: demographics, diagnosis, medical history, pathological report, laboratory test results, surgical records, medication therapies and follow-up information after previous care. All the tables were standardized following the pre-defined format (supplementary information in the GitHub repository).

Data curation results and the layout of PHKGs

PHKGs with individual medical information were created during the data curation process from the data tables. For each patient, the data curation process was executed separately inside each table source to form a Source Health Knowledge Graph (SHKG). The PHKG can be considered as a longitudinal record integrating all 12 SHKGs. As graph outputs, the RDF triples with the Universal Resource Locator (URL), indicating the location of each data element in the PHKG, were stored and maintained in a GraphDB database. Figure 5 shows an example of the visualization of RDF triples from the data mapping for clinical TNM stage. (Examples of SHKGs with RDF triples can be found in the GitHub repository.)

Data query example and the layout of breast cancer registry

We found 72 data items consistent with the Dutch national breast cancer registries, where 14 items were identified as essential elements for a federated registry across three medical sites (see Table 1). A full SPARQL query was developed to extract targeted data from the PHKGs, to form a local breast cancer registry. Crucial data categories in our site-specific registry can include demographics, medical history, diagnosis, pathological report, surgery information, chemotherapy and other systemic treatments, radiotherapy and relevant side effects, tumor screening, and follow up. Full registry information and demo SPARQL queries are available in the GitHub repository. Figure 6 shows a partial example of a SPARQL query for the TNM stage.

Table 1
Core data items of the federated breast cancer registry

Categories	Variable description
Demographics	Birth date
Demographics	Gender
Demographics	Body mass index (BMI)
Diagnosis	Clinical T-stage
Diagnosis	Clinical N-stage
Diagnosis	Clinical M-stage
Diagnosis	Diagnosis date
Diagnosis	Age at diagnosis
Diagnosis	Laterality
Medical history	Menopausal status at diagnosis
Pathological report	Number of involved nodes
Therapy	Neoadjuvant chemotherapy
Therapy	Adjuvant chemotherapy
Therapy	Adjuvant radiotherapy

Discussion

In our study, the first version of the AI-powered virtual assistant prototype (AIDAVA G1 prototype) was successfully developed, and the workflow for semi-automated data curation and publishing was tested by using real patient data in breast cancer. A SPHN-based reference ontology was developed as the target format for the SHKGs and related PHKGs. Being the curation outputs, the longitudinal PHKGs include comprehensive personal health information that can be transformed into multiple outputs, including an international patient summary format, through a full SPARQL query across a single PHKG. Additionally, breast cancer registries can be developed and maintained via another SPARQL query across multiple PHKGs. Based on the study results, this workflow has the potential to realize effective heterogenous personal health data harmonization and publishing. Our work can lead to other benefits. It creates interoperable and reusable personal health data to provide reliable sources for the development of generalizable AI models for diagnosis, prediction and decision-making. Furthermore, our workflow showcases how to integrate multiple AI techniques in a systematic way for effective health data curation. For instance, NLP was applied to extract the text-based data from the electronic health records and match the data with specific medical terms or codes included in the reference ontology. Deep learning technique was implemented in the development of a translation module as well as an AIDAVA-specific data imputation tool to identify and impute missing information, such as missing relationships to other nodes

within a health knowledge graph, therefore enhancing the completeness of PHKGs. Collectively, these components demonstrate a cohesive, AI-enabled approach that streamlines health data processing and supports scalable, high-quality data creation.

Compared to previous studies for the development of knowledge graphs in breast cancer, our study used previously collected patient data. The patients had an interactive experience to ingest and curate their own data through the HITL process at the front end of the AIDAVA VA prototype. Then the patients saw their medical data structured as an international patient summary that follows HL7 FHIR format[35]. This increases the patient's awareness as well as autonomy regarding their own data. Clinicians and statisticians can also use this workflow to standardize the registry data into interoperable formats and improve the quality of registries while reducing time and costs of data maintenance, which now requires manual efforts. Furthermore, this workflow provides the potential to develop federated cancer registries across regions for large-scale analyses while maintaining local data privacy and security. The clinicians can interact with both the local and federated cancer registries, then find target information to support further management and research in breast cancer care.

Our workflow of data curation and harmonization is fully language agnostic, enabling straightforward reuse across different institutions. However, this language-agnostic capability is contingent upon the availability of robust natural language processing (NLP) tools for each target language. The framework further supports the development of longitudinal patient-centered knowledge graphs (PHKGs) for additional diseases, as the ontology and pipeline used for PHKG construction can be continually refined and adapted across different therapeutic areas.

Despite promising results toward a potential workflow for semi-automated data curation and knowledge graph construction, our study has some limitations. Firstly, the data came from a limited number of patients and certain missing data led to insufficient capability to validate the whole workflow. Since it will take a long time to recruit patients and collect the data, synthetic test data based on real-world health data can be similarly used to test the AI-powered workflow during the development stage for the AIDAVA G2 prototype[36]. Furthermore, according to the study results from the G1 prototype, we need to focus on improving the integration of different tools and smoothing the whole workflow for a more effective curation process. Also, the PHKG structure, including the ontology, is mainly based on two therapeutic areas and the patient data from the hospitals. If we include large-scale data in the future with diverse patient populations, we will need to consider how to develop a comprehensive PHKG that can be self-updated continuously. Relevant concerns can be the method to define and create new concepts and relationships according to the new data, the adjustment and expansion of current ontology, the way to impute missing data, and the adapted validation process to ensure mapping quality.

Additionally, in our study, we used SHACL, which has been recommended for the validation of knowledge graphs composed by RDF[37]. As a language for expressing constraints on RDF-based graphs, SHACL defines the structure of RDF triples, such as required properties, necessary relationships, acceptable datatypes and allowed values for certain data nodes. SHACL validation ensures adherence to the predefined schema of a knowledge graph, thereby enhancing data quality and interoperability, and paving

the way for the development of downstream applications. However, SHACL validation focuses more on structural constraints rather than semantics. This process may not detect inaccurate results according to real-world knowledge and will ignore semantic errors that can be important in the interpretation of health data. For instance, neoadjuvant chemotherapy is usually given before the surgery to shrink the tumor and reduce metastases. The date of neoadjuvant chemotherapy should not be later than the surgery in the treatment plan of breast cancer. However, a wrongly recorded date of neoadjuvant therapy in a health knowledge graph may not be detected by SHACL. Furthermore, SHACL does not explicitly define the semantics of recursive shapes in medical knowledge graphs[38]. Recursive shapes are SHACL constraints that refer to themselves directly or indirectly, such as “Patient A has a diagnosis of Breast Cancer in the subtype of Hormone Receptor Positive Cancer that is a subtype of Cancer”. As health data often contain complex, hierarchical, or interconnected relationships, this limitation leads to ambiguity in validation results. Therefore, semantic-aware SHACL validation models, or complementary validation approaches with semantic check and reasoning support can be solutions to improve data quality after the construction of medical knowledge graphs. A difficulty in semantic checking is identifying data elements with variable descriptions or typos. NLP and LLMs can be effective tools for key information extraction, reasoning and semantic match, which can make up the deficits of traditional SHACL validation.

In the future, we will need an upgraded G2 prototype to further explore the potential of this AI-powered health data curation workflow, as well as more measures to evaluate the quality of PHKGs, which can give us more evidence about the feasibility of the AIDAVA virtual assistant.

Conclusion

The proposed workflow is feasible for semi-automated patient data harmonization and knowledge graph construction in breast cancer. Furthermore, it provides a potential solution to enhance medical data interoperability and facilitate the maintenance of clinical registries in breast cancer. It could be extended to broader clinical registries.

Declarations

Ethical approval disclosure

This study was approved by the Institutional Review Board of Maastro Clinic, Maastricht University (approval number: P0720) and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants prior to inclusion in the study. All identifiable patient information extracted from the Maastro Clinic’s EHR system were maintained in the local server and only used in the scope of the AIDAVA project to ensure the data security and privacy.

Author contributions

Study design: P.K., I.Z., and R.C. Acquisition of clinical data: P.K., R.M., and A.B. Data processing: W.L., E.E., O.K., K.S., and T.P. Data interpretation: W.L. Manuscript drafting: W.L. Manuscript revision: W.L., R.M., P.K.,

A.D., I.Z., A.B., A.G., C.W., M.K., M.M., R.C., K.S., S.B., and the AIDAVA consortium. Final approval of manuscript: all authors.

Acknowledgments

The authors appreciate the generous support from all the members of the AIDAVA consortium with their expertise. A special thanks goes to the Breast Cancer and Cardiovascular Disease patient consultants from the ECPC, Data for Patients, EHN patient organizations, North Estonia Medical Centre (NEMC) and Medical University of Graz (MUG); they keep providing the study team with patient-centric insights and solid support.

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Funding

The author(s) discloses support for the research of this work from the European Union’s Horizon Europe research and innovation programme [grant agreement No. 101057062 (AIDAVA)]. The work of the Swiss Partner (MIDATA) received funding by the Swiss State Secretariat for Education, Research and Innovation (SBFI), subvention contract 22.00093, REF-1131-52104. Parts of the work have received funding from the Austrian Science Fund (FWF) [grant number: P-32554 explainable Artificial Intelligence]. The funders played no role in study design, data collection, analysis and interpretation of data, or the writing of this manuscript.

Competing interests statement

I.Z. is employed by b!loba. K.S., T.P., and S.B. are employed by Ontotext AD. The remaining authors declare no competing interests.

Data availability

The data generated and/or analyzed during the current study and its supplementary information files are available in the AIDAVA GitHub repository: https://github.com/MaastrichtU-CDS/AIDAVA-BC

Code availability

The scripts for data mapping outputs and information queries are available in the AIDAVA GitHub repository: https://github.com/MaastrichtU-CDS/AIDAVA-BC

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Figures

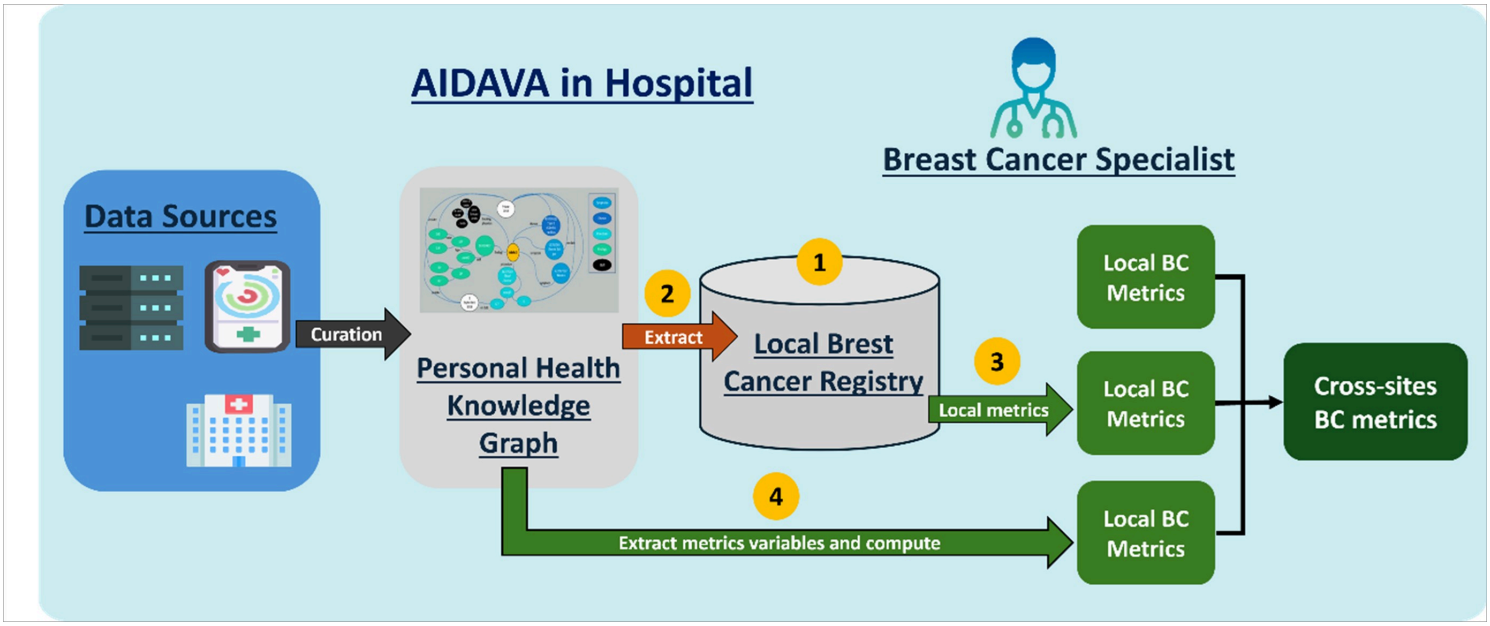

Figure 1

Local data flows in the AIDAVA project for breast cancer

Step 1 - 2. Multi-sourced personal health data from a hospital can be ingested in the AIDAVA virtual assistant prototype, then curated and harmonized in the form of PHKGs. Targeted data elements can be queried from the PHKGs to develop local BC registries.

Step 3. Local metrics can be computed from each medical site and integrated to evaluate the performance.

Step 4. The metrics can also be directly extracted and computed from PHKGs.

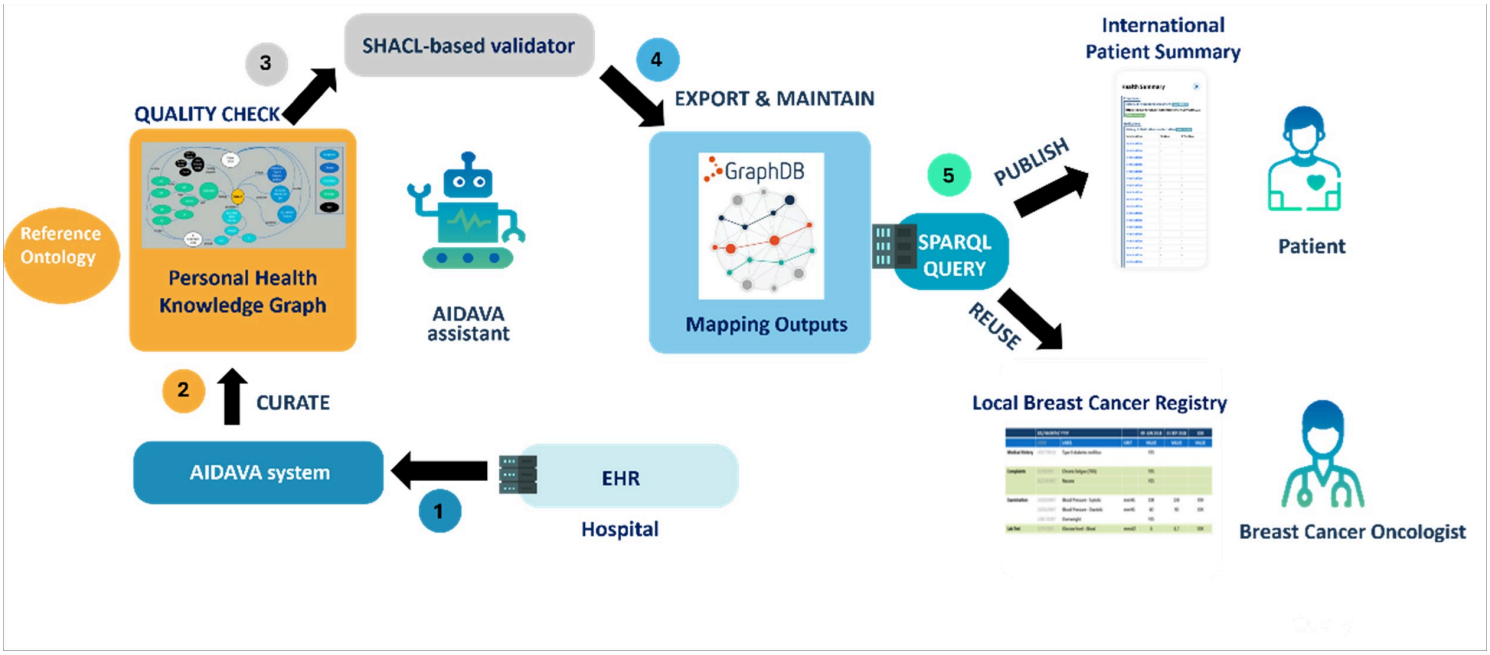

Figure 2

The site-specific process of patient data curation and publishing via AIDAVA

This figure shows the life cycle of electronic health record (EHR) data in the AIDAVA workflow: 1. patient data transferred from the hospital EHR systems to the AIDAVA prototype; 2. data standardization, transformation and PHKG creation; 3. quality enhancement; 4. validated PHKG exported and maintained in a graph database; 5.information query to publish the data into international patient summary and reuse the data to develop a local breast cancer registry. EHR: electronic health records, SPARQL: SPARQL Protocol and RDF Query Language.

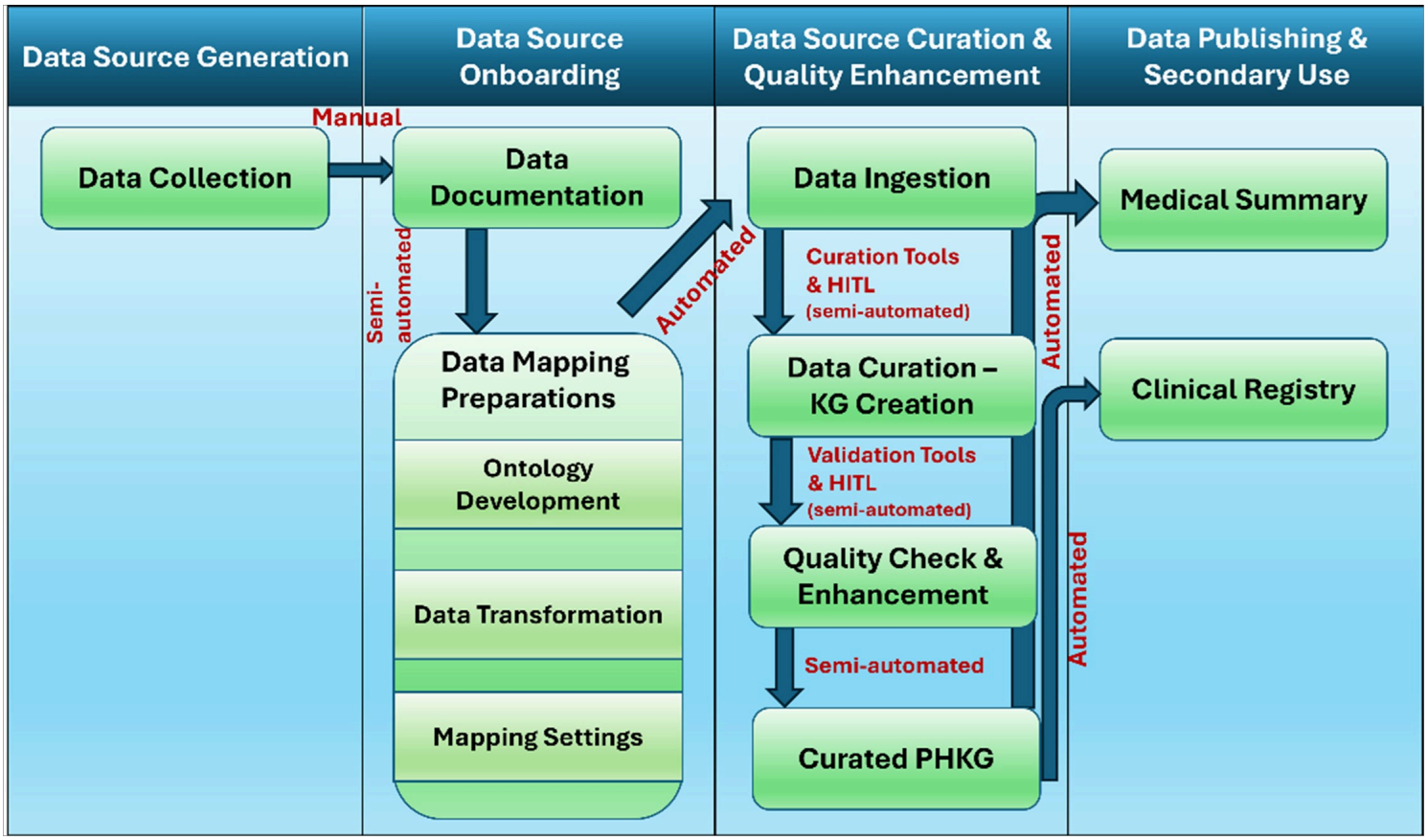

Figure 3

Data harmonization and reuse in AIDAVA

The semi-automated AIDAVA workflow includes four critical phases. KG: knowledge graph; HITL: human in the loop; PHKG: personal health knowledge graph.

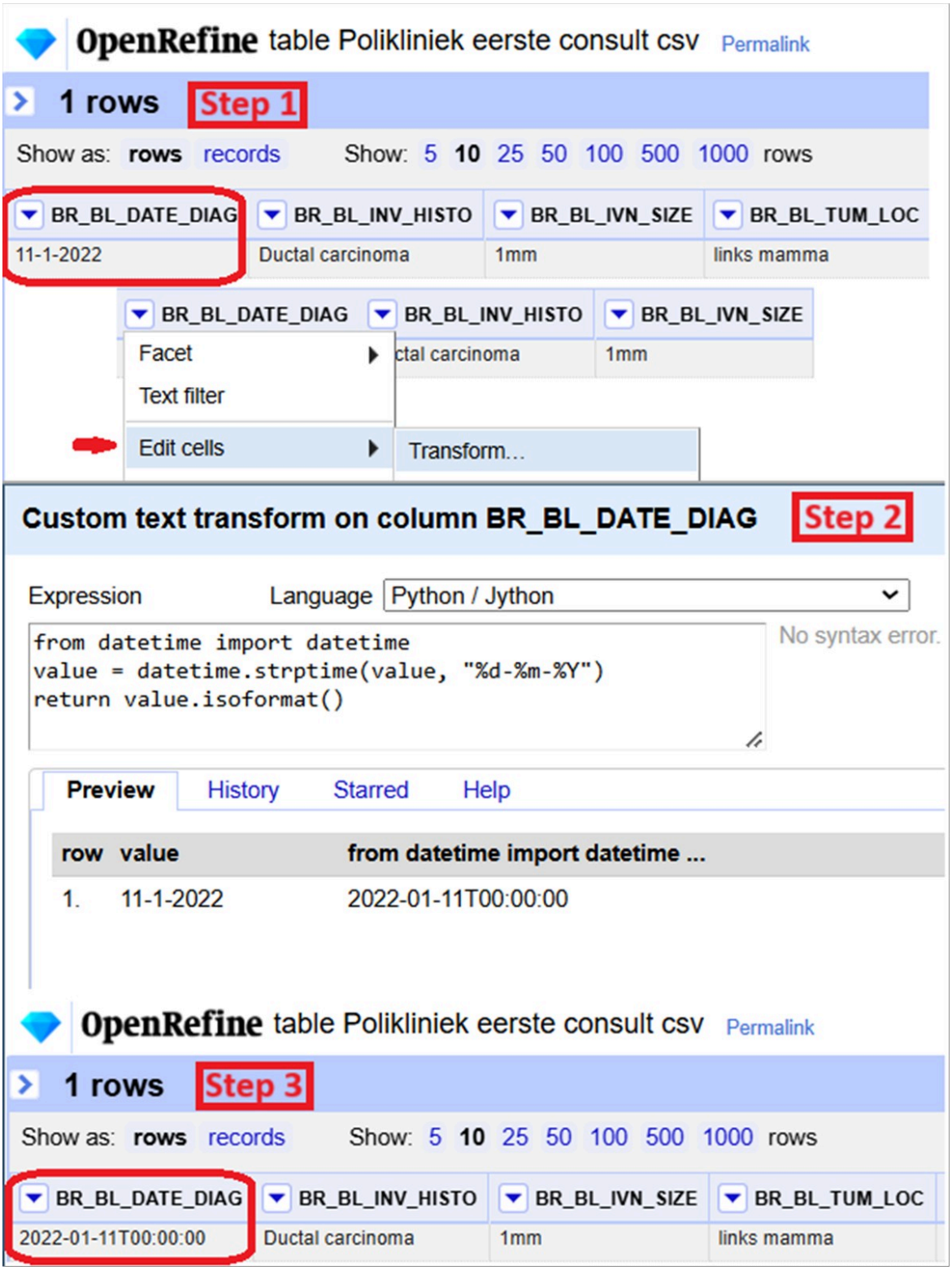

Figure 4

Time data transformation by using integrated OpenRefine

It shows how the datetime data can be standardized in the ISO 8601 date format by using an integrated tool in the AIDAVA system.

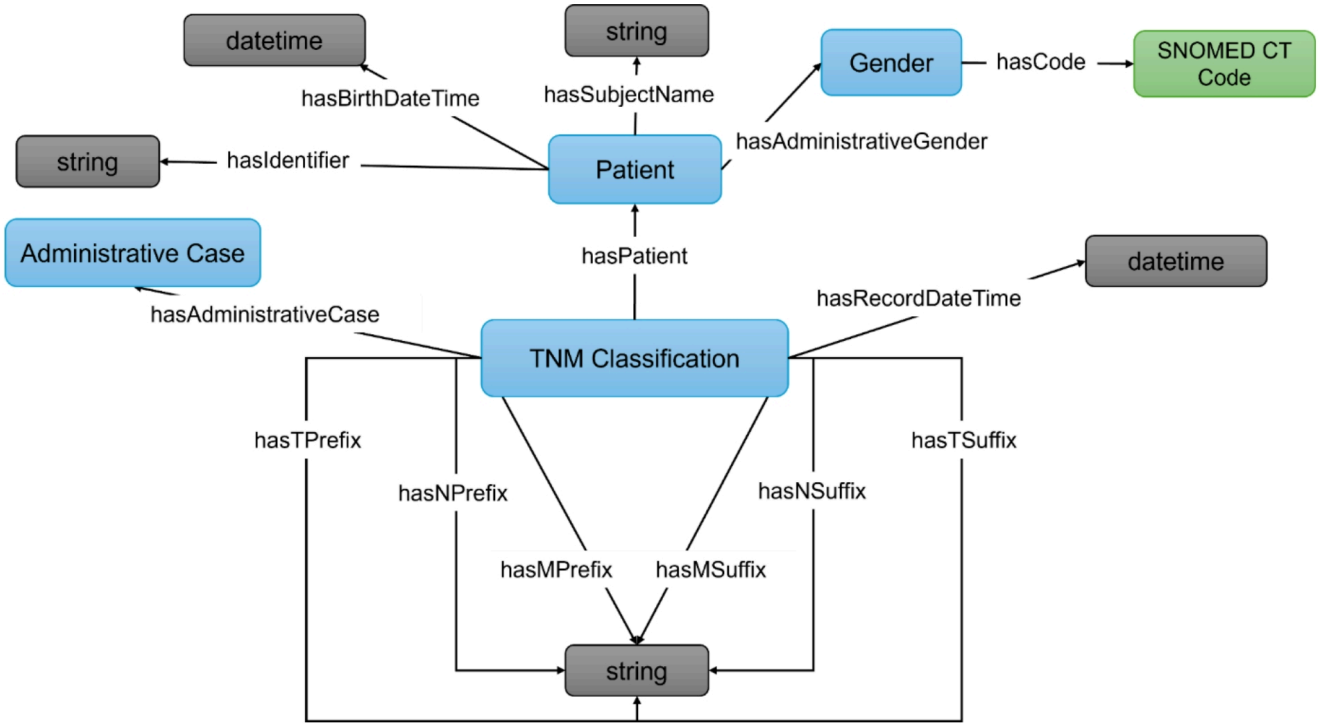

Figure 5

Mapping example for TNM stage

Blue data nodes represent different classes defined in the reference ontology, with grey nodes for data types and a green node for a relevant code in SNOMED CT terminology.

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX sphn-loinc: <https://biomedit.ch/rdf/sphn-resource/loinc/>
PREFIX atc: <https://www.whocc.no/atc_ddd_index/?code=>
PREFIX anno: <http://aidava.eu/annotations/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX aidava: <https://biomedit.ch/rdf/sphn-ontology/AIDAVA/>
PREFIX sphn: <https://biomedit.ch/rdf/sphn-ontology/sphn#>
PREFIX skos: <http://www.w3.org/2004/02/skos/core#>
PREFIX snomed: <http://snomed.info/id/>

SELECT DISTINCT ?patient
?tnm_stage_value
(IF(BOUND(?t_prefix), IF(BOUND(?t_suffix), CONCAT(?t_prefix, ?t_suffix), ?t_prefix), IF(BOUND(?t_suffix), ?t_suffix, "")) AS ?t_stage)
(IF(BOUND(?n_prefix), IF(BOUND(?n_suffix), CONCAT(?n_prefix, ?n_suffix), ?n_prefix), IF(BOUND(?n_suffix), ?n_suffix, "")) AS ?n_stage)
(IF(BOUND(?m_prefix), IF(BOUND(?m_suffix), CONCAT(?m_prefix, ?m_suffix), ?m_prefix), IF(BOUND(?m_suffix), ?m_suffix, "")) AS ?m_stage)
WHERE {
    # TNM Stage
    OPTIONAL {
        {
            SELECT *
            WHERE {
                GRAPH <http://rdf.aidava.eu/metadata#metagraph>
                {
                    ?g <http://rdf.aidava.eu/metadata#belongsTo> ?patient .
                }
                GRAPH ?g {
                    ?tnm_stage rdf:type sphn:TNMClassification .
                    OPTIONAL {?tnm_stage sphn:hasValue ?tnm_stage_value .}
                    OPTIONAL {?tnm_stage sphn:hasTPrefix ?t_prefix .}
                    OPTIONAL {?tnm_stage sphn:hasTSuffix ?t_suffix .}
                      OPTIONAL {?tnm_stage sphn:hasNPrefix ?n_prefix .}
                    OPTIONAL {?tnm_stage sphn:hasNSuffix ?n_suffix .}
                    OPTIONAL {?tnm_stage sphn:hasMPrefix ?m_prefix .}
                    OPTIONAL {?tnm_stage sphn:hasMSuffix ?m_suffix .}
                }
            }
        }
    }
}
```

Figure 6

SPARQL query for TNM stage